Unit 0.01: ACT Math Diagnostic & Pacing Strategy Drill

Q1. For the imaginary unit i , where i^2 = -1 , which of the following complex numbers is equal to 5 - 3i - 2 - 8i ?

[x] A: 3 + 5i
[ ] B: 3 - 11i
[ ] C: 7 - 11i
[ ] D: 3 - 5i

Solution: To subtract complex numbers, distribute the negative sign to both terms inside the second set of parentheses: 5 - 3i - 2 - 8i = 5 - 3i - 2 + 8i Next, combine the real parts and the imaginary parts separately: Real parts: 5 - 2 = 3 Imaginary parts: -3 + 8i = 5i Combining these yields 3 + 5i . Common Misconceptions: - 3 - 11i occurs if the negative sign is not distributed to -8i , yielding -3i - 8i = -11i . - 7 - 11i occurs if the real parts are added 5 + 2 = 7 instead of subtracted. - 3 - 5i occurs from an arithmetic sign error when combining -3 + 8 .

Q2. The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?

[ ] A: 14%
[ ] B: 28%
[x] C: 35%
[ ] D: 42%

Solution: To find the percentage of total hours logged by Maya: 1. Find the total number of volunteer hours logged by all four students: text{Total hours} = 8 + 14 + 6 + 12 = 40 2. Determine the fraction of hours Maya contributed: frac{text{Maya's hours}}{text{Total hours}} = frac{14}{40} 3. Simplify the fraction and convert to a percentage: frac{14}{40} = frac{7}{20} = frac{35}{100} = 35% Common Errors: - Choosing 14% mistakenly treats the raw number of hours as a percentage without dividing by the total. - Choosing 28% corresponds to calculating frac{14}{50} instead of using the actual total of 40 hours.

Q3. In triangle ABC shown below, the measure of angle A is 52^circ and the measure of angle B is 68^circ . What is the value of x ?

[ ] A: 58
[x] B: 60
[ ] C: 70
[ ] D: 120

Solution: The sum of the interior angle measures in any triangle is 180^circ . Set up the equation for the sum of the angles in triangle ABC : angle A + angle B + angle C = 180^circ Substitute the given values into the equation: 52^circ + 68^circ + x^circ = 180^circ Combine the known angle measures: 120^circ + x^circ = 180^circ Subtract 120^circ from both sides: x = 180 - 120 = 60 - A common error is calculating 52 + 68 = 120 and stopping there, which gives 120 . - Another mistake is an arithmetic error such as 180 - 110 = 70 .

Q4. In the right triangle triangle ABC shown below, the right angle is at vertex C. The lengths of the sides are given in inches: AC = 12, BC = 5, and AB = 13. What is the value of cos B?

[x] A: frac{5}{13}
[ ] B: frac{12}{13}
[ ] C: frac{5}{12}
[ ] D: frac{12}{5}

Solution: In any right triangle, the cosine of an acute angle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse: cos B = frac{text{length of side adjacent to } B}{text{length of hypotenuse}} = frac{BC}{AB} From the given triangle: - The side adjacent to angle B is BC = 5. - The hypotenuse is AB = 13. Therefore, cos B = frac{5}{13}. Common Misconceptions: - frac{12}{13} represents sin B = frac{text{opposite}}{text{hypotenuse}} = frac{AC}{AB} or cos A. - frac{5}{12} represents frac{text{adjacent}}{text{opposite}} = cot B or tan A. - frac{12}{5} represents tan B = frac{text{opposite}}{text{adjacent}} = frac{AC}{BC}.

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Q5. For the imaginary unit i , where i^2 = -1 , which of the following complex numbers is equal to 5 - 3i - 2 - 8i ?

[x] A: 3 + 5i
[ ] B: 3 - 11i
[ ] C: 7 - 11i
[ ] D: 3 - 5i

Solution: To subtract complex numbers, distribute the negative sign to both terms inside the second set of parentheses: 5 - 3i - 2 - 8i = 5 - 3i - 2 + 8i Next, combine the real parts and the imaginary parts separately: Real parts: 5 - 2 = 3 Imaginary parts: -3 + 8i = 5i Combining these yields 3 + 5i . Common Misconceptions: - 3 - 11i occurs if the negative sign is not distributed to -8i , yielding -3i - 8i = -11i . - 7 - 11i occurs if the real parts are added 5 + 2 = 7 instead of subtracted. - 3 - 5i occurs from an arithmetic sign error when combining -3 + 8 .

Q6. Which of the following expressions is equivalent to 5x^2 - 2x + 7 - 3x^2 - 6x - 4 ?

[ ] A: 2x^2 - 8x + 3
[ ] B: 2x^2 - 8x + 11
[ ] C: 2x^2 + 4x + 3
[x] D: 2x^2 + 4x + 11

Solution: To simplify the expression, distribute the negative sign across each term in the second set of parentheses: -3x^2 - 6x - 4 = -3x^2 + 6x + 4 Now rewrite the entire expression and combine like terms: 5x^2 - 2x + 7 - 3x^2 + 6x + 4 Group terms with the same power of x : - Quadratic terms: 5x^2 - 3x^2 = 2x^2 - Linear terms: -2x + 6x = 4x - Constant terms: 7 + 4 = 11 Combining these yields 2x^2 + 4x + 11 . Common misconceptions: - Failing to distribute the negative sign to -6x yields -2x - 6x = -8x , producing the distractor 2x^2 - 8x + 11 . - Failing to distribute the negative sign to -4 yields 7 - 4 = 3 , producing the distractor 2x^2 + 4x + 3 . - Failing to distribute the negative sign to both terms yields 2x^2 - 8x + 3 .

Q7. In the standard x, y coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints -3, 8 and 5, -2?

[x] A: 1, 3
[ ] B: 4, -5
[ ] C: 2, 6
[ ] D: 3, 1

Solution: To find the midpoint x_m, y_m of a line segment with endpoints x_1, y_1 and x_2, y_2, use the midpoint formula: x_m = frac{x_1 + x_2}{2}, quad y_m = frac{y_1 + y_2}{2} Substitute the given coordinates -3, 8 and 5, -2: x_m = frac{-3 + 5}{2} = frac{2}{2} = 1 y_m = frac{8 + -2}{2} = frac{6}{2} = 3 Thus, the midpoint is 1, 3. Common errors: - Finding the difference instead of the average yields leftfrac{5 - -3}{2}, frac{-2 - 8}{2}right = 4, -5. - Forgetting to divide the sum by 2 yields 2, 6. - Reversing the x- and y-coordinates yields 3, 1.

Q8. If fx = 3x^2 - 5x + 4 , what is the value of f-2 ?

[ ] A: -18
[ ] B: 2
[ ] C: 6
[x] D: 26

Solution: To evaluate f-2 , substitute -2 for x in the function rule: f-2 = 3-2^2 - 5-2 + 4 . First, evaluate the exponent: -2^2 = 4 . Next, perform the multiplications: 34 = 12 and -5-2 = +10 . Finally, sum the terms: 12 + 10 + 4 = 26 . Common errors include treating -2^2 as -4 , which yields -12 + 10 + 4 = 2 , or failing to apply the negative sign to the middle term, giving 12 - 10 + 4 = 6 .